

Wenjie Luo

Transport Planning and Civil Engineering | Graduating Jan 2027

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PROFILE

Combine site coordination with SUMO traffic studies and SAP2000 modeling. Use measured calibration, independent checks and option comparisons to produce engineering recommendations and reports.

EDUCATION

- National University of Singapore | MSc Civil Engineering**Aug 2025 - Jan 2027
- Transport focus; research in transportation information and control.
- Chang'an University | BEng Transportation Engineering**Sep 2021 - Jun 2025
- GPA 3.16/5.0; approximately top 15% of the major.

EXPERIENCE

- Far East Facade | Project Engineer, Engineering Department**Jun 2026 - Present
- Assist the project manager with 15 façade material types and 5 suppliers, tracking testing, deliveries and installation; contribute to contractor meetings and monthly reporting.
 - Participate in subcontractor discussions on equipment and staffing; inspect installation progress and quality, identify deviations and follow up on corrective actions.
- Expeditors | Supply Chain Assistant**Dec 2024 - Apr 2025
- Coordinated carriers, clients and customs brokers on shipping documentation; issued and reviewed declarations and improved existing Excel workflows and data checks.

PROJECTS

- CE5001 | Multimodal Flood-Resilience Evaluation**Jan - Apr 2026
- Integrated DEM terrain, roads, travel demand and inundation into a SUMO dissertation pipeline: approximately 3,500 Python lines across 10 modules, covering 793 bus routes and 5,201 stops.
 - Completed 200 paired comparisons of rules, persona-augmented rules and models. On dense networks, the rule-based plan cut relevant travel time by 62.2%, informing deployment recommendations.
- CE5203 | AYE Weaving-Bottleneck Study**Mar - Apr 2026
- Owned YOLOv11 video counts and SUMO modeling in a four-person team; analyzed 43 peak and 60 off-peak minutes, extracting flow and speed at four measurement locations.
 - Original coursework compared 12 VSL, ramp-metering and combined scenarios. Reported reductions were 22.7% in peak network time loss and 7.7% in average weaving delay.
- SAP2000 | Reinforced-Concrete Frame Analysis**Sep - Oct 2025
- Led a coursework model of a three-storey frame using C30 concrete, HRB400 reinforcement and approximately 120 elements; evaluated five load combinations.
 - Performed static, modal and seismic analyses, checked internal forces against calculations and addressed two overstress locations; delivered 15 result plots and a technical report.

SKILLS

Engineering: SUMO/TraCI, YOLOv11 traffic counts, SAP2000, ABAQUS, ArcGIS, AutoCAD
Methods: Python, experimental design, simulation calibration, structural checks, material planning